



PRS Pricing Model — Test Results

Automated Test Documentation for Model Governance

MKM Research Labs

Version 1.0 — 06-April-2026

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1 Test Results — PRS Pricing Model (MKM-PR-001)

Tests run on **06 April 2026 at 07:49:50**.

Table 1: Test Results Summary — PRS Pricing Model (MKM-PR-001)

Total Tests	57
Passed	57
Failed	0
Skipped	0
Duration	7.70s

Table 2: Detailed Test Results — PRS Pricing Model

Test Description	Acceptance Criteria	Result	Time
1Y from Feb 2026 = last Friday May 2027.	<code>mat == last_friday_of_month(2027, 5)</code>	Pass	0.000s
3Y from Feb 2026 = last Friday May 2029.	<code>mat == last_friday_of_month(2029, 5);</code> <code>mat.month == 5 (+1 more)</code>	Pass	0.000s
Jan 2027 uses Aug 2026 roll → Nov 2029 maturity.	<code>mat == last_friday_of_month(2029, 11)</code>	Pass	0.000s
3Y from Sep 2026 = last Friday Nov 2029 (Aug roll).	<code>mat == last_friday_of_month(2029, 11);</code> <code>mat.month == 11</code>	Pass	0.000s
5Y from Feb 2026 = last Friday May 2031.	<code>mat == last_friday_of_month(2031, 5)</code>	Pass	0.000s
5Y from Sep 2026 = last Friday Nov 2031.	<code>mat == last_friday_of_month(2031, 11)</code>	Pass	0.000s
All maturities should fall on a Friday.	<code>mat.weekday() == 4</code>	Pass	0.000s
3Y maturity jumps from May to Nov at the Aug roll.	<code>may_mat.month == 5; nov_mat.month == 11 (+1 more)</code>	Pass	0.000s
Aug onwards uses Aug roll.	<code>roll == last_friday_of_month(2026, 8);</code> <code>roll.month == 8</code>	Pass	0.000s
December uses the same year's Aug roll.	<code>roll.month == 8; roll.year == 2026</code>	Pass	0.000s
Feb-Jul uses Feb roll.	<code>roll == last_friday_of_month(2026, 2);</code> <code>roll.month == 2</code>	Pass	0.000s
January uses the previous year's Aug roll.	<code>roll == last_friday_of_month(2026, 8);</code> <code>roll.month == 8 (+1 more)</code>	Pass	0.000s
July is the last month of the Feb roll period.	<code>roll.month == 2; roll.year == 2026</code>	Pass	0.000s
March is still in the Feb roll period.	<code>roll.month == 2; roll.year == 2026</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
compute_maturity_date with no ref_date uses today.	<code>isinstance(mat, date); mat.weekday() == 4</code>	Pass	0.000s
current_roll_date(None) uses today and returns a valid date.	<code>isinstance(roll, date); roll.weekday() == 4</code>	Pass	0.000s
maturity_table with no ref_date uses today.	<code>len(table) == 5; all(isinstance(e['maturity_date'], date) for e in table)</code>	Pass	0.000s
Empty propertyts dir	<code>stats["total_properties"] == 0; stats["properties_processed"] == 0</code>	Pass	0.011s
Generator should still work without gauge hazard curves (empty basis).	<code>stats["properties_processed"] == 3; stats["properties_skipped"] == 0 (+1 more)</code>	Pass	0.272s
No propertyts dir	—	Pass	0.000s
Spread is flat term structure	<code>all(s == spreads[0] for s in spreads)</code>	Pass	0.273s
Spread proportional to floods	<code>spread3 > spread1 > spread2</code>	Pass	0.278s
Zero floods zero spread	<code>spreads[0] == expected</code>	Pass	0.318s
Returns severe event count	<code>gen._get_gauge_severe_count(gauge_hc) == 7</code>	Pass	0.000s
Returns zero for empty dict	<code>gen._get_gauge_severe_count({}) == 0</code>	Pass	0.000s
Returns zero when missing	<code>gen._get_gauge_severe_count(gauge_hc) == 0</code>	Pass	0.000s
Aug 2026: 31st is Monday, last Friday = 28th.	<code>last_friday_of_month(2026, 8) == date(2026, 8, 28)</code>	Pass	0.000s
Feb 2026: 28th is Saturday, last Friday = 27th.	<code>last_friday_of_month(2026, 2) == date(2026, 2, 27)</code>	Pass	0.000s
May 2029: 31st is Thursday, last Friday = 25th.	<code>last_friday_of_month(2029, 5) == date(2029, 5, 25)</code>	Pass	0.000s
Nov 2029: 30th is Friday.	<code>last_friday_of_month(2029, 11) == date(2029, 11, 30)</code>	Pass	0.000s
Result should always be a Friday (weekday=4).	<code>d.weekday() == 4</code>	Pass	0.000s
Actual years should be strictly increasing.	<code>years[i] > years[i - 1]</code>	Pass	0.000s
All maturities should be in May for the Feb roll.	<code>e['maturity_date'].month == 5</code>	Pass	0.000s
All maturities should be in November for the Aug roll.	<code>e['maturity_date'].month == 11</code>	Pass	0.000s
Maturity string should be formatted as 'DD Mon YYYY'.	<code>len(e['maturity_str']) > 8; 'May' in e['maturity_str'] or 'Nov' in e['maturity_str']</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
Default table has 5 entries (1Y through 5Y).	<code>len(table) == 5</code>	Pass	0.000s
Tenors should be 1, 2, 3, 4, 5.	<code>tenors == [1, 2, 3, 4, 5]</code>	Pass	0.000s
Basis calculation	<code>len(nearest) > 0; ng0["gauge_id"] == "GAUGE-001" (+5 more)</code>	Pass	0.315s
Depth thresholds	<code>"severe" in thresholds; "annual_probability" in thresholds["severe"] (+1 more)</code>	Pass	0.296s
Generate produces output	<code>output_path.exists(); "metadata" in data (+2 more)</code>	Pass	0.278s
Gev params always none	<code>pc["gev_params"] is None</code>	Pass	0.296s
PROP-003 has 10 floods, nearest gauge has 8 severe events.	<code>prop3["summary"]["flood_transmission_rate"] > 0</code>	Pass	0.305s
Output structure	<code>"PROP-001" in curves; "PROP-002" in curves (+13 more)</code>	Pass	0.308s
Properties processed	<code>stats["properties_processed"] == 3; stats["properties_skipped"] == 0</code>	Pass	0.279s
With event count pricing, term structure spread is flat (storms independent).	<code>s == spreads[0]</code>	Pass	0.313s
Term structure	<code>ts["tenors"] == TENORS; "severe" in ts (+2 more)</code>	Pass	0.290s
After attach_spread_decomposition, propertyhc.json has decomposition.	<code>"spread_decomposition" in pc; "gauge_spread_bps" in sd (+5 more)</code>	Pass	0.825s
At least one effect per path must be non-zero when gauge != property.	<code>df["distance_effect_bps"] != 0.0 or df["elevation_effect...; ef["elevation_effect_bps"] != 0.0 or ef["distance_effect...]</code>	Pass	0.870s
Both decomposition paths should arrive at the property spread.	<code>abs(path1 - prop) < 0.1; abs(path2 - prop) < 0.1</code>	Pass	1.057s
With shd/she files present, decomposition values must be non-zero.	<code>sd["property_spread_bps"] != 0.0; sd["shd_spread_bps"] != 0.0 (+1 more)</code>	Pass	0.830s
Without shd/she files, decomposition still works (zeros).	<code>n > 0; sd["shd_spread_bps"] == 0.0 (+1 more)</code>	Pass	0.282s
<code>_get_5yr_spread</code> must return index 4 when available.	<code>PropertyHazardCurveGenerator._get_5yr_spread() == 50.0</code>	Pass	0.000s
<code>_get_5yr_spread</code> must return 0.0 for empty/missing data.	<code>PropertyHazardCurveGenerator._get_5yr_spread() == 0.0; PropertyHazardCurveGenerator._get_5yr_spread(... (+1 more)</code>	Pass	0.000s

Test Description	Acceptance Criteria	Result	Time
<code>_get_5yr_spread</code> must work with specified threshold.	<code>PropertyHazardCurveGenerator._get_5yr_spread(pc, 'severe'...;</code> <code>PropertyHazardCurveGenerator._get_5yr_spread(pc, 'any_flo...</code>	Pass	0.000s
All tenors should round-trip through maturity and back.	<code>tenor_from_maturity(mat, ref) == t</code>	Pass	0.000s
A date not in the table returns the closest tenor.	<code>result in [1, 2, 3, 4, 5]</code>	Pass	0.000s
Exact maturity date returns correct tenor.	<code>tenor_from_maturity(mat, date(2026, 2, 24)) == 3</code>	Pass	0.000s

1.1 Test Document History

Date	Version	Description	Author
05-Mar-2026	1.0	Initial test suite (30 tests)	D. Kelly
07-Mar-2026	1.1	57 test(s) added; 30 test(s) removed (total: 57)	D. Kelly
07-Mar-2026	1.2	30 test(s) added (total: 87)	D. Kelly
18-Mar-2026	1.3	4 test(s) added (total: 91)	D. Kelly
27-Mar-2026	1.4	30 test(s) removed (total: 61)	D. Kelly
30-Mar-2026	1.5	3 test(s) added; 12 test(s) removed (total: 52)	D. Kelly
01-Apr-2026	1.6	4 test(s) added (total: 56)	D. Kelly
03-Apr-2026	1.7	3 test(s) added; 2 test(s) removed (total: 57)	D. Kelly
06-Apr-2026	1.8	9 test(s) added; 9 test(s) removed (total: 57)	D. Kelly